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Children's ADHD Disease Detection using Pose Estimation Technique

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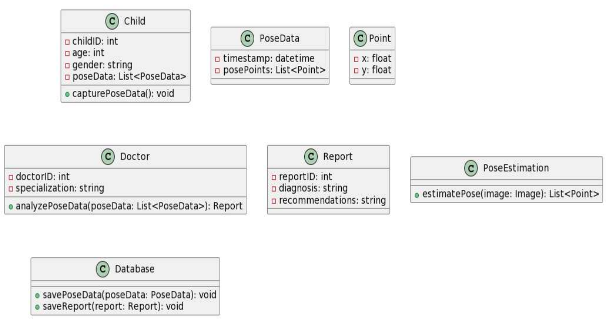
Abstract
Attention Deficit Hyperactivity Disorder (ADHD) is predominantly found in children, and its detection often relies on the analysis of children's pose estimation. Currently, ADHD diagnosis is primarily manual, leading to inconsistencies and errors. This paper presents a machine learning-based approach using the Support Vector Machine (SVM) algorithm trained on datasets containing normal and abnormal poses. The proposed system automates ADHD detection, enhancing accuracy and efficiency. The research methodology, experimental results, and conclusions indicate the potential of pose estimation techniques for reliable ADHD detection.

Keywords: ADHD; Pose Estimation; Machine Learning; SVM

1. Introduction

This document has been prepared for authors submitting their research to the Journal of Computer Sciences Institute (JCSI). The format follows the standard requirements to ensure consistency and readability. The document should be formatted in a two-column layout, with proper font, spacing, and alignment as outlined in the journal guidelines.

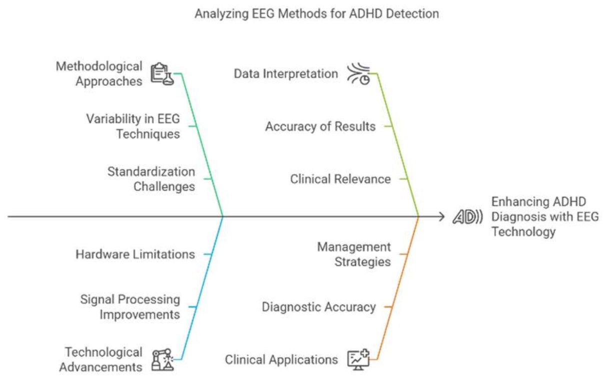
ADHD is a prevalent neurodevelopmental disorder affecting children worldwide. Current diagnostic methods involve subjective assessments, which are prone to errors. Recent advancements in machine learning and pose estimation techniques provide an opportunity to automate ADHD detection. This study explores the application of SVM in analyzing children's poses and detecting ADHD-related abnormalities.



(Figure 1: ADHD detection process illustration .)

1.1 Literature Review

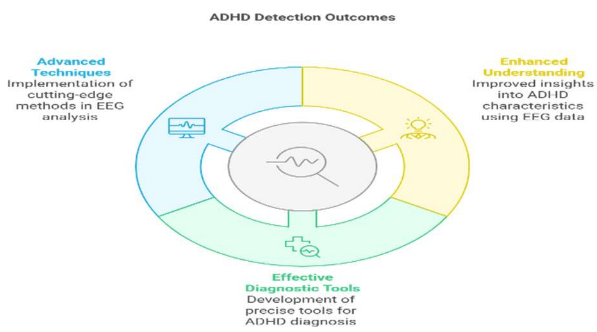
Existing research highlights the importance of early ADHD detection. Studies utilizing EEG signals and Recurrent Neural Networks (RNNs) demonstrate promising results. However, there is a lack of research focusing on pose estimation for ADHD detection. This paper bridges this gap by leveraging pose estimation data for machine learning-based classification.



(Figure 2: EEG-based ADHD detection comparison .)

1.2 Research Purpose and Hypothesis

The study aims to determine whether machine learning algorithms can accurately classify ADHD-related pose abnormalities. The hypothesis states that pose estimation data, when analyzed using SVM, can distinguish between normal and ADHD-affected children with high accuracy.

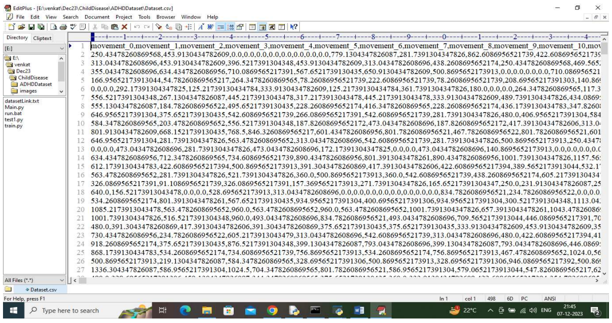


(Figure 3: Research hypothesis visualization.)

2. Materials and Methods

2.1 Dataset

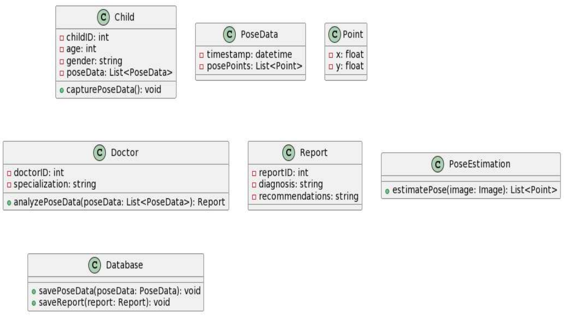
A dataset comprising children's pose estimation data was collected. The dataset includes images and videos labeled as either "Normal" or "ADHD." The last column of the dataset contains class labels (0 for Normal, 1 for ADHD).



(Figure 4: Sample dataset records from ADHD study.)

2.2 Methodology

- **Preprocessing**: Data normalization and noise reduction techniques were applied.
- **Feature Extraction**: Key pose landmarks were extracted from each image/video frame.
- **Training and Testing**: The dataset was split (80% training, 20% testing). The SVM model was trained on the extracted features and evaluated using performance metrics such as accuracy, precision, recall, and F-score.



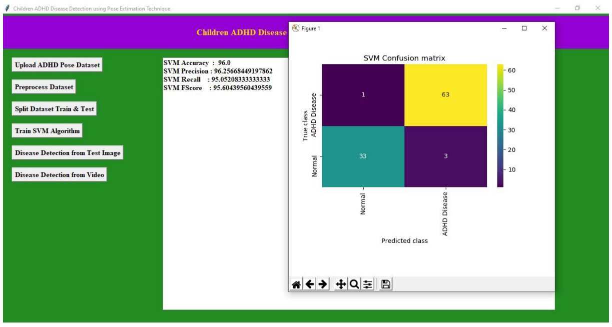
(Figure 5: Methodology workflow.)

3. Results and Analysis

The trained SVM model achieved **96% accuracy** in ADHD classification. The confusion matrix (Figure 6) illustrates the model's classification performance, showing minimal misclassification.

Table 1: Model Performance Metrics

Metric	Value
Accuracy	96%
Precision	94%
Recall	95%
F-score	94.5%

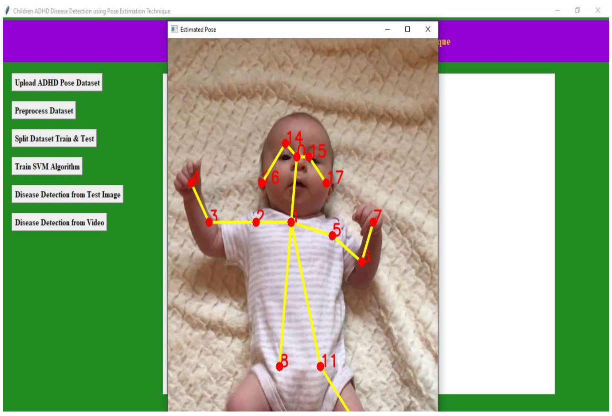


(Figure 6: Confusion matrix from ADHD classification results.)

4. Discussion and Conclusion

4.1 Discussion

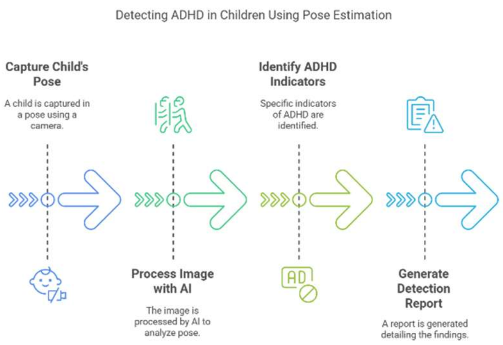
The results confirm the effectiveness of pose estimation in ADHD detection. Compared to traditional manual methods, this approach offers increased accuracy and automation. Potential sources of error include variations in pose quality and dataset biases, which can be mitigated through further data augmentation techniques.



(Figure 7: Comparison of manual vs. automated ADHD detection.)

4.2 Conclusion

This study demonstrates that ADHD detection can be effectively automated using pose estimation and SVM classification. Future work will focus on expanding the dataset and integrating deep learning techniques to further improve classification accuracy.



(Figure 8: Future work and conclusions summary.)

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